

FLIGHT BOOKING PREDICTIONS

Data Description

Business Context:

An aviation company offering both domestic and international travel services is looking to improve its customer acquisition strategy. Traditionally, outreach has been done through broad-based methods like telecalling, which are costly and less effective. To modernize their approach, the company has partnered with a social networking platform to leverage customer digital and social behavior for targeted advertising. Since running advertisements on digital platforms is expensive, it is essential to focus marketing efforts only on customers who are most likely to purchase tickets.

Objective:

As a data scientist, you are tasked with building predictive models that estimate the propensity of customers to purchase tickets based on their online behavior.

Two separate models must be developed - one for laptop users and another for mobile users, since device usage influences purchase patterns. Accurate models will help the company minimize advertising costs while maximizing conversions, leading to more efficient campaigns and improved return on investment.

Data Description:

The dataset contains different attributes for potential customers of the aviation company. The detailed data dictionary is given below:

- **UserID:** Unique ID of user
- **Taken_product:** Buy a ticket next month
- **Yearly_avg_view_on_travel_page:** Average yearly views on any travel-related page by user

- **preferred_device:** Through which device user prefer to do login
- **total_likes_on_outstation_checkin_given:** Total number of likes given by a user on out-of-station check-ins in the last year
- **yearly_avg_Outstation_checkins:** Average number of out-of-station check-ins done by the user
- **member_in_family:** Total number of relationships mentioned by the user in the account
- **preferred_location_type:** Preferred type of location for travelling of user
- **Yearly_avg_comment_on_travel_page:** Average yearly comments on any travel-related page by user
- **total_likes_on_outofstation_checkin_received:** Total number of likes received by a user on out-of-station check-ins in the last year
- **week_since_last_outstation_checkin:** Number of weeks since last out-of-station check-in update by user
- **following_company_page:** Whether the customer is following the company page (Yes or No)
- **montly_avg_comment_on_company_page:** Average monthly comments on the company page by user
- **working_flag:** Whether the customer is working or not
- **travelling_network_rating:** Does the user have close friends who also like travelling? 1 is high and 4 is lowest
- **Adult_flag:** Whether the customer is an adult or not
- **Daily_Avg_mins_spend_on_traveling_page:** Average time spent on the company page by user on a daily basis